

# HealthConnect Testing & Validation Report

## Week 7 Technical Deliverable — Programmatic Dashboard Verification & Reassessment

|                             |                             |                                     |                         |
|-----------------------------|-----------------------------|-------------------------------------|-------------------------|
| DATASET<br>Appointment Data | TOTAL RECORDS<br>5,000 Rows | ENVIRONMENT<br>Python venv (Pandas) | STATUS<br>PASSED (100%) |
|-----------------------------|-----------------------------|-------------------------------------|-------------------------|

### 1. Executive Summary & Objective

This validation report documents programmatic testing, data integrity audits, and segment-level consistency checks performed for the HealthConnect appointment analytics platform. Utilizing Python (pandas), raw records (HealthConnect\_Appointment\_Data.csv) were cross-verified against dashboard metrics.

#### Key Audit Finding

Programmatic calculations confirm 100% agreement with dashboard summary visual cards. Zero rounding discrepancies or logical drift were detected between the database and visual layers.

### 2. Data Quality & Structural Audit

Initial structural inspection flagged specific completeness gaps across secondary attributes. Pre-processing rules were verified to ensure missing attributes do not distort core KPI aggregations:

| Attribute Name        | Missing Count | Impact Level | Validation Action                                    |
|-----------------------|---------------|--------------|------------------------------------------------------|
| reminder_channel      | 1,366         | Moderate     | Isolated nulls for channel effectiveness cross-tabs. |
| distance_to_clinic_km | 90            | Low          | Excluded from distance mean calculations via dropna. |
| waiting_time_minutes  | 60            | Low          | Audited for wait-time outlier distribution.          |

### 3. Comprehensive Testing & Validation Record

| Test ID | Validation Category   | Programmatic Output (Python)                            | Dashboard Output                              | Status |
|---------|-----------------------|---------------------------------------------------------|-----------------------------------------------|--------|
| VAL-01  | Baseline KPI Volume   | 5,000 total records verified exact match                | Total Appointment KPI card matches 5,000      | PASS   |
| VAL-02  | Segmented Consistency | Cross-tabs confirm lead-time trends hold across subsets | Filtered slicer views maintain logical trends | PASS   |
| VAL-03  | Filter Responsiveness | appointment_type = General Consultation (n=3,255)       | Slicer selection dynamically updates visuals  | PASS   |
| VAL-04  | Anomaly Audit         | Zero negative lead times; wait-time outliers flagged    | Data range boundaries enforced correctly      | PASS   |

## 4. Segmented Conclusion Testing

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To ensure core insights are robust, conclusions were tested across operational segments. Segmented cross-tabulations confirm that the elevated no-show risk associated with extended booking lead times remains consistent across major appointment types and age brackets, proving findings are broadly generalizable.

## 5. Analytical Errors, Weaknesses & Limitations

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Transparent evaluation of analytical constraints ensures high technical rigor:

- **Unimputed Missing Data Gaps:** Gaps in `reminder_channel` (1,366), `distance_to_clinic_km` (90), and `waiting_time_minutes` (60) were excluded list-wise rather than imputed, potentially introducing minor attrition bias.
- **Confounding Variables:** Lead-time correlations are treated as observational risk indicators rather than direct causal drivers, as unmeasured factors (e.g., patient health urgency) may influence behavior.
- **Sub-Segment Sample Variance:** Highly granular filter slices yield smaller sample sizes, which can cause percentage volatility in niche demographic brackets.

## 6. Reassessed Strategic Recommendations

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Based on completed validation testing, segment checks, and analytical audits, the following evidence-based operational recommendations are established:

- **Lead-Time-Triggered Intervention Protocols:** Replace static reminder schedules with a tiered, dynamic engagement model. For appointments booked >14 days out, trigger automated check-ins at critical milestones (14 days, 7 days, and 48 hours prior) with interactive confirmation options.
- **Channel Effectiveness Optimization:** Standardize mandatory logging of communication channels during booking and pivot toward omni-channel automated workflows (SMS paired with WhatsApp or voice confirmation) tailored to high-risk service types.
- **Waiting Time & Distance Mitigation:** Establish real-time tracking for clinic wait times and integrate distance/travel factors into scheduling algorithms—offering telehealth alternatives for patients located further from the clinic (>10 km) to reduce transportation barriers.